PROJECT ABSTRACT

Project Title: AI-Based Resume Screening and Candidate Ranking System Using DevSecOps and MLOps

Abstract

Recruitment is a vital organizational process, but manually reviewing large numbers of resumes is slow, repetitive, and prone to inconsistency. This project proposes an AI-Based Resume Screening and Candidate Ranking System that automates the initial stage of recruitment by comparing resumes against job descriptions using natural language processing and machine learning. The application extracts relevant information including skills, education, certifications, and experience from uploaded resumes, preprocesses the text, and computes a relevance score to rank candidates. Recruiters receive an ordered list of applicants along with matched and missing skills, enabling faster and more consistent shortlisting.

The project is developed using Agile Scrum with iterative sprints, user stories, backlog management, and continuous feedback. Git and GitHub are used for version control, while Docker containerizes the application and GitHub Actions provides continuous integration. Kubernetes (Minikube) demonstrates orchestration. Security is integrated throughout the lifecycle using SonarQube for static analysis, Trivy for container vulnerability scanning, OWASP ZAP for dynamic testing, secure coding practices, and environment-based secret management. MLOps practices include dataset versioning using DVC, experiment tracking and model registry using MLflow, and application monitoring using Prometheus and Grafana. Together, these tools create a complete software engineering workflow that is secure, scalable, maintainable, and aligned with industry standards. The proposed solution reduces recruitment time, improves consistency, and provides a practical demonstration of Adaptive Software Engineering concepts.

Problem Statement

Organizations often receive hundreds of applications for a single job opening. Manual screening consumes significant time and may overlook qualified candidates because of human error or inconsistent evaluation. There is a need for an intelligent system that automates resume analysis, objectively compares candidates with job requirements, and supports recruiters with accurate rankings while following modern software engineering, security, and deployment practices.

Objectives

- Automate resume screening.
- Rank candidates based on job relevance.
- Reduce manual effort and recruitment time.
- Demonstrate Agile, DevOps, DevSecOps, and MLOps.
- Deploy a secure and scalable application.

Methodology

The application accepts resumes in PDF/DOCX format, extracts text, cleans and preprocesses the content, converts it into numerical vectors using TF-IDF, and calculates similarity with the job description. The trained model is exposed through a Flask API and integrated with a web interface. Development follows Scrum. Automated testing and CI/CD are implemented using GitHub Actions. Docker and Kubernetes are used for deployment. Security and monitoring tools are integrated to demonstrate an end-to-end engineering workflow.

Expected Outcome

The final system will provide recruiters with a web-based interface to upload resumes and job descriptions, generate candidate rankings, identify missing skills, and improve hiring efficiency. It will also serve as a complete demonstration of Agile development, DevOps automation, DevSecOps security, and MLOps lifecycle management.